At low temperature isotherms, for example, at 13º C as the pressure increases, the volume decreases along AB and is a gas until the point B is reached. At B, a liquid separates along the line BC, both the liquid and gas co-exist and the pressure remains constant. At C, the gas is completely converted into liquid. If the pressure is higher than at C, only the liquid is compressed so, there is no significant change in the volume. The successive isotherms shows similar trend with the shorter flat region. i.e. The volume range in which the liquid and gas coexist becomes shorter. At the temperature of 31.1 ºC the length of the shorter portion is reduced to zero at point P. In other words, the CO_2 gas is liquefied completely at this point. This temperature is known as the liquefaction temperature or critical temperature of CO_2. At this point the pressure is 73 atm. Above this temperature CO_2 remains as a gas at all pressure values. It is then proved that many real gases behave in a similar manner to carbon dioxide.

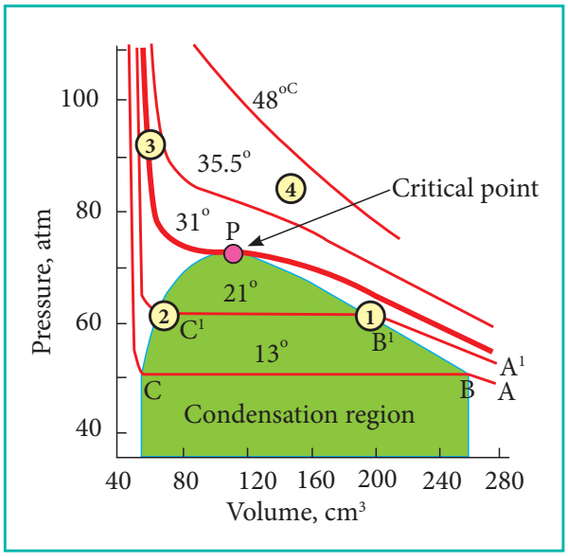

Figure. 6.12 Isotherms of Carbon dioxide at different tempeartures

Though the nature of isotherm remains similar, the critical temperature, the corresponding pressure and volume are characteristics of a particular gas.

Now we can define the critical constants as follows. Critical temperature (T_c) of a gas is defined as the temperature above which it cannot be liquefied even at high pressure. Critical pressure (P_c) of a gas is defined as the minimum pressure required to liquefy 1 mole of a gas at its critical temperature. Critical volume (V_c) is defined as the volume occupied by 1 mole of a gas at its critical temperature and critical pressure. The critical constants of some common gases are given in Table 6.2

Table 6.2 Critical constants of some gases

Name of the Gas	Critical Temperature (T_c) in K	Critical Pressure (P_c) in atm	Critical Volume (V_c) cm^3 mol^{-1}
Helium (He)	5.2	2.26	57.8
Carbon dioxide (CO_2)	304.2	72.9	94.0
Oxygen (O_2)	154.8	50.14	78.0
Nitrogen (N_2)	126.3	33.54	90.1
Hydrogen (H_2)	33.2	12.80	65
Water (H_2O)	647.4	218.3	55.3
Ammonia (NH_3)	405.5	111.3	72.5
Hydrogen Chloride (HCl)	324.7	81.5	81.0
Methane (CH_4)	190.6	45.6	98.7
Ethylene (C_2H_4)	283.1	50.50	124

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